

This set of exercises covers chapters 1-9 of our textbook with the exception of chapter 7 and advanced material. To pass and be allowed to take the final exam, you have to get at least 75% of all points (total 28). You get only one chance, there will be no second round of answers allowed. You must upload a PDF file with your answers and names of all team members to mitt.uib.no. If you write your solutions using pen and paper then scan them in the correct order and make a pdf. Good luck!

Problem 1. For this problem consider Figure 1.

1. Perform Breadth First Search (BFS) starting at node "1". Describe the layer in which each node is discovered.
(1 point)
2. Describe one possible use of the BFS algorithm for analysing social networks.
(1 point)

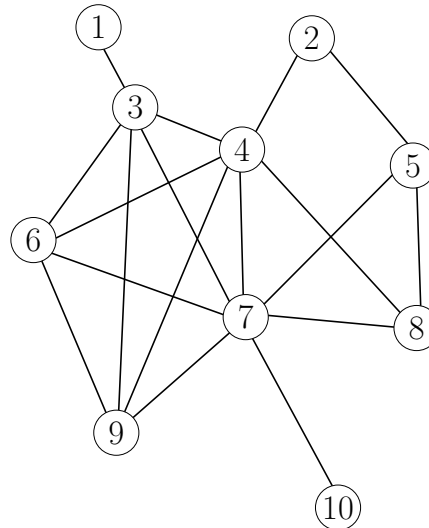


Figure 1: BFS.

Problem 2. Consider Figure 2.

1. List all bridges, and separately all local bridges that are not also bridges, in the network shown below.
(1 point)
2. For the edges that are not local bridges, use the concept of neighborhood overlap to argue which edge comes closest to playing the role of a local bridge.
(1 point)

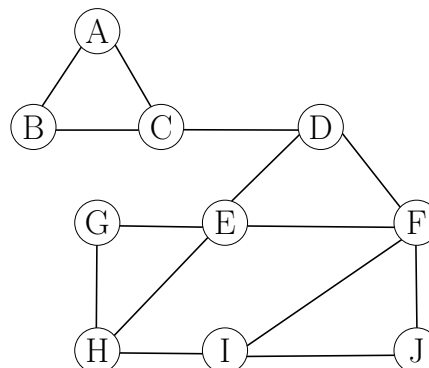


Figure 2: Bridges.

Problem 3.

1. In Figure 3, which nodes satisfy the Strong Triadic Closure property, and which nodes do not?
(1 point)

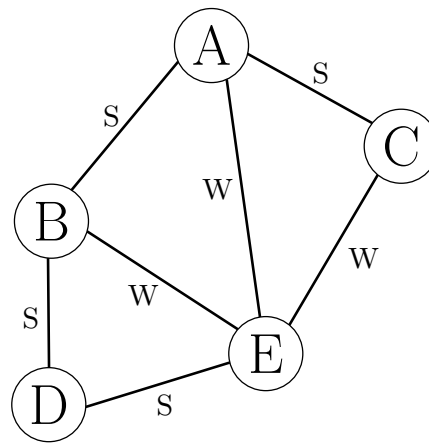


Figure 3: Triadic closure.

2. Explain the connection between Strong Triadic Closure, weak ties, and local bridges.
(1 point)

Problem 4.

1. Explain how focal closure and membership closure in a social affiliation network relate to homophily arising from either selection or social influence.
(1 point)
2. To make Schelling segregation deterministic, impose the following additional rules:
 - (a) After marking unhappy nodes, first move the top-left one, then the remaining nodes in the top row, and continue in this way for each row.
 - (b) When moving a node, move it to the closest available position, where distance is the number of steps up/down plus the number of steps left/right.
 - (c) Perform Schelling segregation on Figure 4 with threshold 3, following these rules.
(1 point)

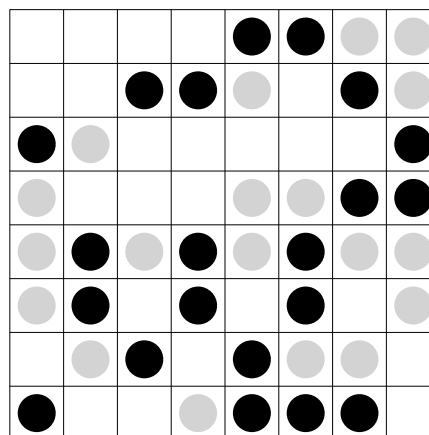


Figure 4: Schelling segregation.

Problem 5. Explain how the Schelling segregation model shows a mutable characteristic becoming correlated with an immutable characteristic, and why the final result is not an integrated pattern even though such a pattern would satisfy each individual.
(1 point)

Problem 6. Consider Figure 5, where every edge is labelled either as a friend (positive) or an enemy (negative) relation.

1. Which triangles are not balanced?
(0.5 point)

2. Which triangles are weakly balanced?
(0.5 point)
3. What is the minimum number of edge changes (from + to − or vice versa) required to make all triangles weakly balanced? Present the changes.
(1 point)
4. For each type of triangle labelled by + and −, explain why that type is more abundant or less abundant in some social setting of your choice.
(1 point)

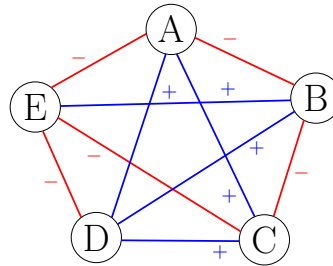


Figure 5: Network with positive and negative edges.

Problem 7. Recall the basic description of the prisoner's dilemma game as presented in the book:

Two individuals, called Karkaroff and Malfoy, are suspected of being involved in a crime together. They are interrogated in separate rooms so that they cannot communicate with each other. Each is given the same two choices by law enforcement personnel: stay silent (choice 1), or confess to their involvement and snitch on the other person (choice 2).

If both stay silent, each is sentenced to 3 years in prison (for lack of conclusive evidence). If one suspect confesses while the other stays silent, the one who confesses receives only 1 year (for cooperating with law enforcement), while the other serves 10 years. If both confess, each serves 5 years.

For each of the following modified versions of the game (in each case, all the information above still applies), construct a bi-matrix representation that you consider an acceptable model of the situation, and make a reasonable prediction for the outcome. State your solution concept clearly and briefly explain your answer.

1. Suppose that both players are completely selfless and each cares only about reducing the other player's prison sentence, with no regard for his own fate.
(0.5 point)
2. Suppose that each player "feels the other's pain" in addition to his own; that is, if Karkaroff serves x years and Malfoy serves y years, each suffers a cost of $(x + y)$.
(0.5 point)
3. Suppose that Malfoy has connections to important people in law enforcement. If both suspects stay silent, Malfoy gets off scot-free (0 years), while Karkaroff gets 2 years. Each suspect selfishly cares about reducing his own prison sentence.
(0.5 point)
4. Which strategies are social-welfare optimizing?
(0.5 point)

Problem 8.

1. Find all pure (nonrandomized) strategy Nash equilibria in the game described by Table 1. Explain your answer.
(1 point)
2. Let p denote the probability that the row player plays U , so that D is played with probability $1 - p$. Let q denote the probability that the column player plays L , so that R is played with probability $1 - q$. Express all probabilities p and q as fractions. Find all mixed-strategy Nash equilibria in the game. Show your calculations.
(1 point)
3. Explain the difference between Pareto optimality and social optimality.
(1 point)

Problem 9. Consider the network shown in Figure 6, with each edge labelled by its delay function. There are 300 travellers going from A to B . Let x be the number of travellers using the path $A-C-B$, and let y be the number using the path $A-D-B$.

Player 1 \ Player 2	Player 2	
	L	R
U	3, 1	7, 7
D	5, 4	6, 2

Table 1: 2-Player game.

1. What is the equilibrium split of traffic?
(1 point)
2. Assume travel cost is expressed in money, i.e. the monetary value of people's travel time. How would the equilibrium traffic split change if a 0.1-unit toll were imposed on edge $A-C$?
(1 point)
3. Suppose an edge from C to D with cost 0 is added. Does this create an instance of the Braess Paradox?
(1 point)

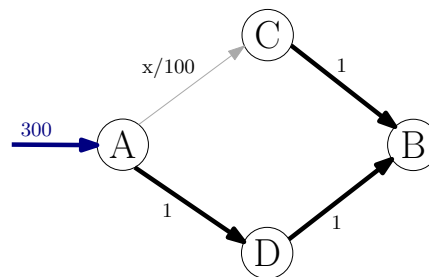


Figure 6: Traffic network.

Problem 10.

1. Define ascending-bid and descending-bid auctions when a single item is being sold.
(1 point)
2. Define first-price sealed-bid and second-price sealed-bid auctions when a single item is being sold.
(1 point)

Problem 11. We end with several coding exercises. The task is to analyse graphs that have been downloaded. The Python package `networkx` will be used within Jupyter Notebook. On MittUiB, there is a folder called Coding Assignment. Download this folder and read the PDF file called coding-setup, and use obligatory-assignment-coding.ipynb. Then answer the following questions.

1. Consider the undirected graphs `0.edges` and `414.edges` from a dataset containing various Facebook friends lists, available for download from MittUiB.
 - (a) What are the sizes of the connected components of these graphs?
(1 point)
 - (b) Do the sizes of the connected components reflect what you would expect to find? Explain.
(1 point)
2. Consider the graph `links.tsv`, containing links between Wikipedia pages, both as a directed graph and as an undirected graph.
 - (a) Does the directed or undirected case best reflect the underlying data, i.e. links between Wikipedia pages? Explain.
(1 point)
 - (b) Can the diameter ever be larger in an undirected version than in the directed version of `links.tsv`? Does this hold for every graph where we compare the directed and undirected versions? Explain.
(1 point)
 - (c) In both cases, what are the diameters of these graphs? Which are the 10 nodes with the highest degrees, and what are their degrees?
(1 point)
 - (d) In both cases, delete the 300 nodes with the highest degrees. What are the diameters now?
(1 point)